

Midterm 1: Virtualization

Lecture 2: Ali

Questions that many people answered wrong are highlighted.

Part 1: Operating systems true/false - 3 questions

For True fill in A, for False fill in B

1. Examples of resources that the OS must manage include CPU, memory, and disk.

True - there are many others but these are examples

2. An operating system can be defined as software that makes hardware into a useful form for applications.

True - there are other definitions, but this is a valid one

3. The operating system is always running, even when user applications are running.

False - it stops running when a user program runs and starts running only after a system call, trap or interrupt

Part 2: CPU virtualization true/false - 15 questions

4. On a timer interrupt, the OS is responsible for transitioning from user to kernel mode.

False - the hardware does this

5. With MLFQ, after running tasks in one queue, the scheduler starts running tasks in the next higher-priority queue.

False - either it will keep running them if they haven't all used their allotment, or it will run tasks in the next *lower* priority queue.

6. With a MLFQ scheduler, jobs run to completion as long as there is not a higher priority job.

False: jobs within a queue run with RR and when they use up their allotment they will be moved to a lower-priority queue

7. If all jobs arrive at the same point in time, a SJF and an STCF scheduler will behave the same.

True: when jobs arrive their time-to-completion is the same as their burst length.

8. An STCF scheduler cannot cause jobs to starve.

False: if lots of short jobs arrive long jobs may never run

9. An RR scheduler tends to increase average response time as the time-slice is decreased.
False - shorter time-slices decrease the average response time.
10. Decreasing the time-slice of an RR scheduler is likely to increase the overhead imposed by scheduling.
True - there will be more context switches and more cache and TLB pollution
11. On a system call, the hardware consults the system call table to find the address of the system call implementation
False - the hardware consults the trap table and the OS uses the system call table.
12. It is unsafe to let user-mode code enable and disable interrupts.
True - this would allow user code to use 100 of the machine and be unkillable
13. With cooperative multitasking, context switches only happen when a program calls yield().
False - they can also happen during any system call
14. Return code of fork() is always 0 for successful execution.
False - for the parent process fork() returns the child PID
15. A child process created using a fork() system call will start its execution at its main function.
False - it starts executing at the return address from fork()
16. A modern OS virtualizes a single CPU with time-sharing.
True - it allow multiple processes to take turns using the CPU using a timer interrupt for preemptive scheduling
17. When an I/O operation completes, the previously blocked process moves into the RUNNING state.
False - it moves to the READY state
18. Whenever a process is descheduled by the scheduler, the process is moved to the BLOCKED state.
False - it is moved to the READY state

Part 3: CPU virtualization multiple choice - 10 questions

19. Which of these tasks does fork **not** do?

- a. Create a new page table
- b. Create a new process control block
- c. Load new code: fork() copies a process and exec() loads new code**
- d. Move a process onto the ready queue

20. Which of the following are both present in processes but not programs

- a. Stack & registers: program has data and code**
- b. Heap & data
- c. Code & registers
- d. Data & code

21. What does a processor do if a process executes a privileged instruction in user mode?

- a. The processor traps: the operating system runs and can terminate the process**
- b. The instruction is ignored
- c. The computer reboots

22. The MLFQ scheduler we studied always starts processes at the highest priority level (runs first). Suppose we change the rules so that processes always start at a middle priority. What rule makes sense for moving processes from that starting priority to a higher priority queue?

- a. They use up their entire time slice
- b. They yield the CPU before finishing their time slice: this shows that the time slice is too short and then could run at a higher priority**
- c. They make a system call: unless the system call blocks, the scheduler just treats it as part of the CPU burst
- d. A & B
- e. B & C popular wrong answer

23. What is the purpose of the Boost mechanism with MLFQ scheduling?

- a. **To ensure short-running jobs do not monopolize the CPU: because short running jobs get priority, if there are many of them longer-running jobs may not get to execute.**
- b. To ensure long-running jobs do not monopolize the CPU
- c. To ensure all processes fairly share the CPU
- d. To minimize average turnaround time

24. You are designing a word processor and want to make sure it responds quickly when a key is struck. Which scheduling goal are you most interested in?

- a. Turnaround time
- b. **Response time: interactive processes care about when they start running to respond quickly**
- c. Fairness
- d. No starvation

25. What happens when using a FIFO scheduler and a long-running task enters the ready queue?

- a. **Head-of-line blocking: this happens when you have to wait behind a long-running task and it slows you down**
- b. Starvation
- c. Preemption
- d. Turnaround

26. Where are register contents stored during a context switch?

- a. User stack and kernel stack
- b. User stack and PCB
- c. **Kernel stack and PCB: user-mode general purpose registers are stored on the kernel stack and kernel-mode general purpose registers are stored in the PCB**
- d. Only the PCB

27. How many tables are consulted by hardware and software to find function addresses during a system call made using the `int` instruction?

- a. 0
- b. 1
- c. **2: the trap table/interrupt vector and the system call table**
- d. 3
- e. 4

28. Which of these operations is not **privileged**?

- a. I/O operations
- b. Changing state of the MMU
- c. **Saving and restoring registers and program counter: these are just normal move instructions**
- d. Programming the hardware timer

Part 4: CPU Virtualization calculations - 7 questions

For the next 3 questions, 3 jobs arrive at different times with characteristics in the table at right.

29. Which scheduler minimizes the completion time of the entire workload?

- a. FIFO
- b. RR
- c. SJF
- d. STCF

Job	Arrival Time	CPU burst
A	0	10
B	5	8
C	12	2

- e. **They all complete at the same time. Schedulers typically don't affect the completion time of the whole workload; just order of tasks within a fixed time.**

30. Given a STCF scheduler, what is the response time for job B?

- a. **5 seconds: A runs from 0:10, B arrives at 5 and starts executing at 10, so response time is 5**
- b. 7 seconds
- c. 10 seconds
- d. 20 seconds
- e. None of the above

31. Given a STCF scheduler, what is the response time for job C?

- a. **0 seconds: it arrives at time 12 and because its time-to-completion (2) is less then the time-to-completion of B (6), it runs immediately**
- b. 2 seconds
- c. 12 seconds
- d. 14 seconds
- e. None of the above

For the next 4 questions, three jobs arrive at approximately the same time, but Job A arrives slightly before Job B, and Job B arrives slightly before job C. Job A requires 2 sec of CPU, Job B is 8 secs, and Job C is 7 secs. Assume a time-slice of 1 sec.

Job	Arrival time	CPU Burst
A	0	2
B	0 just after A	8
C	0 just after B	7

32. Given a FIFO scheduler, what is the turnaround time of job B?

- a. 0 seconds
- b. 2 seconds
- c. 8 seconds
- d. **10 seconds - A runs from time 0-2, then B from 2-10, and turnaround time is how long it is from B's arrival to completion.**
- e. None of the above

33. Given a FIFO scheduler, what is the average response time of the three jobs?

- a. 1 second
- b. 2 seconds
- c. **4 seconds: A has response time 0, B has 2, and C has 10; $(0+2+10)/3 = 4$**
- d. 9.67 seconds
- e. None of the above

34. Given a RR scheduler, what is the turnaround time of job B?

- a. 1 second
- b. 4 seconds
- c. 16 seconds
- d. **17 seconds: the schedule will be ABCABCBCBCBCB, so B finishes last at time 17**
- e. None of the above

35. Given a SJF scheduler, what is the turnaround time of job B?

- a. 2 seconds
- b. 9 seconds
- c. 16 seconds
- d. **17 seconds: the order is A[0:2],C[2:9],B[9:17]**
- e. None of the above

Part 5: Memory Virtualization true/false - 15 questions

For True fill in A, for False fill in B

36. Given a fixed page size, the size of a linear page table doubles when the address space size doubles.
True: you need 1 PTE for each page, so if you have double the address space with the same size page you have double the pages
37. On a TLB miss, a new page in physical memory is allocated by the OS.
False: the CPU will look for the page in the pagetable
38. A page fault occurs if the present bit is clear (equals 0) in a PTE needed for a memory access.
True - it means there isn't a valid translation in the page table
39. A TLB miss is usually slower to handle than a page fault.
False - a TLB miss is usually handled in hardware and is much faster than a page fault
40. Paging approaches suffer from internal fragmentation, which grows worse as the size of a page grows.
True: internal fragmentation is space inside an allocation that isn't needed/used, and with larger pages there is more of this.
41. Given a constant number of bits in a virtual address, the size of a linear page table increases with larger pages.
False: with larger pages and the same size address space, fewer pages are needed so fewer PTEs are needed so page tables are smaller
42. If the valid bit is clear (equals 0) for a virtual page needed for a memory access, the running process is likely to be killed by the OS.
True: this means the address is outside of a valid memory region and will cause a segmentation fault, which usually leads to the process being killed
43. When a system is thrashing, system CPU utilization goes up.
False: CPU utilization goes down because the system is busy doing I/O for page faults
44. The working set of a program gets larger when you have more memory.
False: the working set of a program is independent of how much memory you have - it relates only to how much memory the program accesses in a given amount of time.
45. The TLB hit rate generally increases with larger page sizes.
True: the TLB coverage is higher, so more accesses are likely to be in the TLB
46. Given a large set of data (with, say, a million entries), TLB performance for a linked list is better than an array.
False: arrays have better spatial locality
47. To translate and read a memory address, N references are required if we use a multi-level page table with N levels.
The translation takes N and the read takes 1 for N+1

48. Two different address spaces (that do not share any memory) must contain different valid VPNs from one another.

False - every address spaces can use the same VPNs

49. With segmentation, the address space of each process must be allocated contiguously in physical memory.

False: each segment must be contiguous, both the segments themselves can be distributed to different places in physical memory

50. With pure segmentation (and no other support), fetching and executing an instruction that performs a store from a register to memory will involve exactly two memory references.

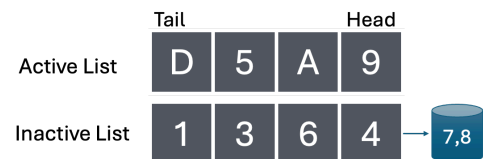
True: the segment table is stored in the MMU and doesn't require memory accesses

Part 6: Memory Virtualization multiple choice - 10 questions

51. What happens if a page's reference bit is set to 1 when the Clock Algorithm scans it?

- a. The page is immediately replaced.
- b. The reference bit is cleared, and the hand moves to the next page.**
- c. The page is moved to the front of the queue.
- d. The algorithm skips the page and moves to the next one without modifying the reference bit.

Given the current state of the Second Chance algorithm shown, answer the following questions.



52. Which is the contents of the active list if page 3 is referenced?

- a. D 5 A 9
- b. D 5 A 3
- c. 3 D 5 A: the system moves 9 to the inactive list**
- d. 3 5 A 9

53. Which page is swapped out to disk if page E is referenced?

Page 4: page 4 is at the head of the inactive list so it is the next one to be evicted.

Page 9

Page 1

No swapping

54. In the x86-64 architecture, the virtual address is 48 bits. In order to support 2 MB pages, how many bits are needed for the page offset, and how many levels does the multi-level page table require? The PTE size is 8 bytes and page tables use 4KB pages..

- a. 21 bits - 3 levels: 2MB = 2^21 so the offset needs 21 bits. With 4KB pages and 8 byte PTEs, a page can hold 512 = 2^9 PTEs. So, each page table uses 9 bits for**

indexing. With 48 VA bits - 21 offset bits = 27 bits for VPN. 27 bits for VPN/9 bits per page table level = 3 levels

- b. 21 bits - 4 levels
- c. 20 bits - 3 levels
- d. 20 bits - 4 levels

55. With Second Chance page replacement, the operating system will choose which page to replace?

- a. The least-recently used page in the process doing a page fault
- b. The least-recently used page across all processes
- c. **A not-recently used page in the process doing a page fault: second chance is a local policy that looks at the pages in a process. It can't determine the precise least-frequently used page and so it events a page that hasn't been used recently.**
- d. A not-recently used page across all processes?

56. Where is *z located in address space?

```
int main()
{
    int *z = malloc(sizeof(int));
    *z = 5;
    return 0;
}
```

- a. **Heap: z contains an address on the heap returned from malloc(), so *z is at that address in the heap**
- b. Stack
- c. Code
- d. No allocation is needed.

The next **two** questions use this information. Assume a system contains three pages (or frames) of physical memory, each of which is originally empty. The access stream is:
ABAC **D** ABCDABC (left to right).

57. Which page is replaced by the OPT page replacement policy when page D is accessed for the first time?

- a. A
- b. B
- c. **C: it isn't used until after A and B**
- d. No replacement needed.

58. Which page is replaced by the LRU page replacement policy when page D is accessed for the first time?

- a. A
- b. **B: A and C are used more recently than B**
- c. C
- d. No replacement needed.

59. Where is the address of the page table directory stored in a multi-level page table when a process is in running state?
- Memory popular wrong answer
 - Disk
 - Registers: the page table directory is stored in the page table base register while a process is running. Note: it is still in the PCB, but this is not where it is used.**
 - Cache
60. When an array is accessed sequentially, the TLB can speed up address translation because of:
- Spatial locality: the program accesses many addresses on the same page in a row**
 - Temporal locality
 - Fewer translations needed
 - Both a and b

Part 7: Memory Virtualization calculations - 10 questions

A system with a linear page table has the following characteristics: Virtual addresses have 15 bits, the page size is 4KB. PTE (page table entry) is 4 bytes. Answer the following questions. Simplified view of the page table:

VPN	PPN
0	0xaa
1	0x89
2	0xab
3	0xac
4	invalid
5	invalid
6	invalid
7	0xbc

61. How many bits does the virtual page number (VPN) have?
- 2
 - 3: 4kb pages is 2^{12} which needs 12 bits; 15 bits in a VA - 12 bits = 3 bits for VPN**
 - 4
 - 5
62. Translate virtual address 0x1abc to physical address.
- 0x891abc
 - 0xaaabc
 - 0x89abc: VPN 0x1 has PPN 0x89**
 - Invalid access
63. How large is the address space?
- 4 KB
 - 16 KB
 - 32 KB: 15 bits in a VA so size is $2^{15} = 32\text{kb}$**
 - 64 KB

Answer the following questions with the following assumptions. 2-level page table, 4 KB pages, 20-bit addresses, and ASID of the current process is 112. The current state of the TLB is as follows:

Valid	Virt	Phys	ASID
0	0x10	0x40	112
1	0x10	0x91	112
1	0x12	0x9E	112
1	0x15	0x55	110

64. How many memory accesses are needed to **translate** the virtual address 0x10101?

- a. **0: page 0x10 is in the second entry with the correct ASID 112 and PPN 0x91**
- b. 1
- c. 2 popular wrong answer
- d. 3

65. How many memory accesses are required to **translate** the virtual address 0x15678?

- a. 0
- b. **2: page 0x15 is in last entry of the TLB but for the wrong ASID (110 instead of 112), so there is a TLB miss needing access to 2 levels of the page table**
- c. 3
- d. Invalid translation

66. What is the physical address of the virtual address 0x10000?

- a. 0x40000
- b. **0x91000: page 0x10 is in the second entry with PPN 0x91**
- c. 0x55000
- d. Cannot be determined using the provided information.

Given that we are using segmented addressing, the Base and Bound register for a process P1 are as follows:

Segment	Base	Bounds	R W
0	0x1000	0xfff	1 0
1	0x2000	0x5ff	1 1
2	0x3000	0x4ff	1 1
3	0x0000	0x000	0 0

Virtual address is 14 bits long and comprises the segment identifier and offset. There are only 4 segments. For this question, ignore that stack grows backwards.

67. Convert 0x0556 Virtual Address to Physical Address.

- a. 0x2556
- b. 0x0556
- c. **0x1556: segment 0 is at base 0x1000**
- d. Segmentation violation

68. Convert 0x2500 Virtual Address to Physical Address.

- a. 0x2500
- b. 0x3500

- c. 0x1500
- d. **Segmentation violation: segment 2 has bounds 0x4ff which is smaller than offset 0x500, so this is an illegal address**

For a system that uses linear page tables with the following characteristics:

- Virtual addresses are 16 bits
- Physical addresses are 20 bits
- Pages are of size 512 bytes

69. The number of bits in the address used for page offset is

- a. 8
- b. **9: 512 bytes = 2^9**
- c. 10
- d. 12

70. The maximum number of virtual pages is

- a. **$2^7 = 128$: there are 16 bit VA - 9 bit offset = 7 bits in the VPN, so there are 2^7 pages**
- b. $2^9 = 512$
- c. $2^{16} = 65536$
- d. $2^{12} = 4096$

